25901321

MAHATMA GANDHI UNIVERSITY,KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEHSC200 - FOOD SCIENCEAND QUALITY CONTROL

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K),Understand(U),Apply(A),Analyse(An),Evaluate(E),Create(C),Skill(S),Interest(I) and Appreciation(Ap)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **10** questions

Each question carries **2 m**arks

1.What is meant by loss of nutrients during cooking? [K]/[CO1]

Reduction of vitamins and minerals in food due to heat, water, and oxidation.

2. Give one example of a commercially available probiotic drink. [K]/[CO3]

Yakult is an example of a commercially available probiotic drink.

3. Mention the causes of food spoilage. [K]/[CO4]

Microorganisms, enzymes, temperature, moisture, and exposure to air.

4. How do the eyes of fresh fish appear? [K]/[CO2]

The eyes of fresh fish appear clear, bright, and slightly bulging.

5. Which alternative protein source is obtained from microorganisms? [K]/[CO3]

Single cell protein (SCP)

6. State the objectives of GMP. [U]/[CO5]

To ensure food safety, quality, and hygiene during production.

7. Which sugar stage is suitable for brittle preparation? [K]/[CO1]

The hard crack stage is suitable for brittle preparation.

8. What is the role of diet in gene expression according to nutrigenomics? [K]/[CO3]

Diet influences gene expression by activating or suppressing specific genes.

9. Give any two examples of intentional adulteration in foods. [K]/[CO4]

Adding water to milk and mixing artificial colors in spices.

10. Name the main parts of an egg. [K]/[CO2]

The shell, albumen (egg white), and yolk.

11. Mention one application of AI in dairy processing. [U]/[CO3]

Quality control and automated milk grading.

12. What is the substrate involved in enzymatic browning?

Phenolic compounds (also called polyphenols). [U]/[CO1]

[2x10=20]

PartB

Short Essay Questions

Answerany**5**questions

Each question carries **4** marks

13. What is homogenization of milk? Explain the process and give two advantages. [K]/[CO2]

Homogenization of milk: It is the process of breaking down large fat globules in milk into smaller, uniformly distributed particles to prevent cream separation.

Process & advantages: Milk is forced under high pressure through narrow openings to reduce fat globule size; this improves digestibility and gives milk a smooth, uniform texture.

14. **Write short notes on two applications of biotechnology in food processing with suitable examples.** [K]/[CO3]

Biotechnology is used in fermentation to produce foods like curd, cheese, and bread using microorganisms such as *Lactobacillus* and yeast.

It is also applied in enzyme technology, for example using amylase in brewing and proteases in meat tenderization.

(2 marks for each application)

15. Describe the structure and main elements of the ISO 22000 food safety management system. [U]/[CO5]

ISO 22000 includes key elements such as interactive communication, system management, prerequisite programs (PRPs), and HACCP principles.

Its structure follows a process approach with continuous improvement through planning, implementation, monitoring, and corrective actions.

(2 marks for structure, 2 marks for elements)

16. **Write short notes on the nutritional value of microgreens.** [K]/[CO1]

Microgreens are rich in vitamins (A, C, E, K), minerals, antioxidants, and bioactive compounds that support immunity and health. They often contain higher nutrient concentrations than mature plants and contribute to disease prevention. (1 mark for each value).

17. **Briefly discuss food-borne viral infections and their transmission through foods.** [U]/[CO4]

Food-borne viral infections are caused by viruses like norovirus and hepatitis A, leading to gastrointestinal illness. (2 marks).

They are transmitted through contaminated food, water, poor hygiene, and infected food handlers. (2 marks)

18. **Explain egg white foam formation – mention the stages of foam formation and any two factors affecting foam stability.** [K]/[CO2]

Egg white foam formation involves stages such as foamy, soft peak, stiff peak, and overbeaten stage due to protein denaturation and air incorporation. (2 marks).

Foam stability is affected by factors like presence of fat (reduces stability) and addition of sugar or acid (improves stability). (2 marks).

19. **Write short notes on anti-nutritional factors in pulses – trypsin inhibitors and lathyrogens (sources, effects and control measures).** [K]/[CO1]

Trypsin inhibitors (in soybeans, pulses) reduce protein digestion, while lathyrogens (in *Lathyrus sativus*) cause neurolathyrism affecting the nervous system. (2 marks)

Control measures include soaking, cooking, fermentation, and avoiding excessive consumption of affected pulses. . (2 marks)

[4x5=20]

Part C

Essay Questions
Answer any**1**
questions
Each question carries **10** marks

20. **Explain the salient features and objectives of the Food Safety and Standards Bill, 2005.** [U]/[CO5]

Salient Features (5 sentences – 5 marks):

1. The Food Safety and Standards Act, 2006 consolidates multiple food laws into a single comprehensive legislation.
2. It establishes the Food Safety and Standards Authority of India to regulate and monitor food safety.
3. It introduces science-based standards for food articles and regulates manufacture, storage, distribution, sale, and import.
4. It emphasizes licensing and registration of food business operators.
5. It provides for penalties, adjudication, and enforcement mechanisms to ensure compliance.

Objectives (5 sentences – 5 marks):

1. To ensure availability of safe and wholesome food for human consumption.
2. To protect public health by preventing food adulteration and contamination.
3. To establish a single reference point for food safety standards.
4. To promote consumer awareness and confidence in food safety.
5. To facilitate trade in food products through uniform standards and regulations.

21. **Discuss the physiological and biochemical changes occurring in fruits during ripening.**

Physiological Changes (5 sentences – 5 marks):

1. Respiration rate increases significantly during ripening (climacteric rise).
2. Fruits soften due to breakdown of cell wall components like pectin.
3. There is a change in colour from green to yellow/red due to chlorophyll degradation and pigment synthesis.
4. Water content and turgor pressure may decrease, affecting texture.
5. Aroma develops due to formation of volatile compounds.

Biochemical Changes (5 sentences – 5 marks):

1. Starch is converted into simple sugars, increasing sweetness.
2. Organic acids decrease, reducing sourness.
3. Enzymes such as amylase, pectinase, and cellulase become more active.
4. Synthesis of pigments like carotenoids and anthocyanins occurs.
5. Production of Ethylene hormone regulates and accelerates ripening processes.

[10x1=10]